WORKED SOLUTION

STEP III 1994, Question 12

Probability and Statistics • First-Step Analysis

STEP III 1994, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q12

The key idea

Look at the games in *pairs*. A lead of 2 can only appear after an even number of games, and after each pair of games one player has served once and the other once, so the serve is back where it started. The match then becomes a simple repeated experiment: each pair is won outright by A, won outright by B, or split, which puts us back at level scores. Standard Tennis is the same thing with the first two pairs treated specially.

Solution

Throughout, write

$$a = p_A p_B, \qquad b = q_A q_B, \qquad s = p_A q_B + q_A p_B.$$

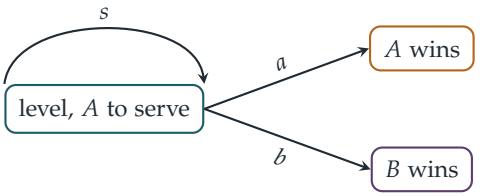
In a pair of consecutive games, one served by each player, a is the probability that A wins both, b that B wins both, and s that they win one each (either A holds serve and B holds serve, or both break). We assume, as the question intends, that the results of different games are independent. These are the only three outcomes, so

$$a + b + s = (p_A + q_A)(p_B + q_B) = 1.$$

Shortened Tennis

The difference in scores changes by ± 1 every game, so it is even after an even number of games and odd after an odd number. A lead of exactly 2 can therefore first occur only after games $2, 4, 6, \ldots$, and if the match is still going after an even number of games the scores are level.

So split the match into pairs (games 1–2, 3–4, ...). Starting from level scores with A to serve, the next pair ends the match in A's favour with probability a, in B's favour with probability b, and with probability s returns us to level scores with A to serve again. The situation is then exactly as at the start, so



$$P_{\text{short}} = a + sP_{\text{short}} \quad \Longrightarrow \quad P_{\text{short}} = \frac{a}{1-s} = \frac{a}{a+b} = \frac{p_A p_B}{p_A p_B + q_A q_B},$$

using $1 - s = a + b$. (Dividing by $1 - s$ is fine unless $s = 1$, the degenerate case where every pair is split and the match never ends.)

If B serves first, each pair consists of a B service game followed by an A service game. The probabilities that A wins both, B wins both, or they split are $p_B p_A$, $q_B q_A$ and $q_B p_A + p_B q_A$: the same a, b, s, because the order of the two games in a pair does not matter. The same argument gives the same answer, so P_{short} does not depend on who serves first.

Decided in 4 games

Nothing can be decided before 4 games have been played. After 4 games (two pairs, A serving first) the difference is even, and the match is decided exactly when the score is 4–0, 3–1, 1–3 or 0–4. In terms of the two pairs:

- 4–0 to A: A wins both pairs, probability $a^2 = p_A^2 p_B^2$;
- 0–4: B wins both pairs, probability $b^2 = q_A^2 q_B^2$;
- 3–1 to A: A wins one pair and the other is split, in either order, probability $2as$;
- 1–3: similarly $2bs$.

Adding,

$$\Pr(\text{decided in 4 games}) = p_A^2 p_B^2 + q_A^2 q_B^2 + 2(p_A p_B + q_A q_B)(p_A q_B + q_A p_B),$$

as required.

Standard Tennis with equal probabilities

If the match is not decided in 4 games, the score is 2–2. This happens with probability $s^2 + 2ab$ (both pairs split, or one pair to each player in either order). From 2–2 the rule "lead by 2" is exactly the Shortened Tennis rule from level scores, and A serves game 5; so from there A wins with probability P_{short}. Hence, in general,

$$P_{\text{stan}} = a^2 + 2as + (s^2 + 2ab)P_{\text{short}}.$$

Now put $p_A = p_B = p$ and $q_A = q_B = q$, so that $a = p^2$, $b = q^2$, $s = 2pq$ and $P_{\text{short}} = \dfrac{p^2}{p^2 + q^2}$. Then

$$P_{\text{stan}} = p^4 + 4p^3 q + \frac{6p^4 q^2}{p^2 + q^2}.$$

Ask yourself Is there a way to compare P_{stan} with P_{short} without expanding everything in powers of p and q?

Yes: unroll Shortened Tennis over the same first four games. Applying $P_{\text{short}} = a + sP_{\text{short}}$ twice,

$$P_{\text{short}} = a + as + s^2 P_{\text{short}}.$$

In words: A wins the first pair; or it is split and A wins the second; or both are split and we start again. Subtracting this from the formula for P_{stan}, the $s^2 P_{\text{short}}$ terms cancel:

$$\begin{aligned} P_{\text{stan}} - P_{\text{short}} &= a^2 + as - a + 2abP_{\text{short}} \\ &= a(a + s - 1) + 2abP_{\text{short}} \\ &= -ab + \frac{2a^2 b}{a+b} \qquad (\text{since } a + s - 1 = -b) \\ &= \frac{ab(a-b)}{a+b}. \end{aligned}$$

This holds for any p_A, p_B. With $a = p^2$ and $b = q^2$,

$$P_{\text{stan}} - P_{\text{short}} = \frac{p^2q^2(p^2 - q^2)}{p^2 + q^2} = \frac{p^2q^2(p-q)(p+q)}{p^2+q^2} = \frac{p^2q^2(p-q)}{p^2+q^2},$$

since $p + q = 1$. This is the required result.

The reason is visible in the working: the two formats differ only when A wins the first pair and B the second, or the other way round. Shortened Tennis awards that to whoever won the first pair; Standard Tennis plays on from 2–2. Both orders have probability ab, so the difference is $ab(2P_{\text{short}} - 1)$, which favours the stronger player.

Checks

- If $p = q = \frac{1}{2}$ the difference is 0, as it must be by symmetry: both probabilities are $\frac{1}{2}$.
- With $p = 0.6$: $P_{\text{short}} = 0.36/0.52 \approx 0.6923$ and $P_{\text{stan}} = 0.1296 + 0.3456 + 0.124416/0.52 \approx 0.7145$, whose difference 0.02215 agrees with $0.0576 \times 0.2/0.52$.
- The probability of a decision in 4 games, $a^2 + b^2 + 2(a+b)s$, and the probability $s^2 + 2ab$ of reaching 2–2 add to $(a+b+s)^2 = 1$, as they should.
- A computer simulation of the actual rules, with $p_A = 0.7$ and $p_B = 0.45$, gives A winning Shortened Tennis about 65.6% of the time whoever serves first, matching $0.315/0.48 = 0.65625$.

Common slips

- Forgetting that 3–1 can arise in two orders (split then win, or win then split), and so missing the factor 2.

- Treating the probability of winning a game as the same for every game: it depends on who serves, which is why grouping into pairs is so useful.
- Writing an infinite geometric series without noting that it is just the equation $P = a + sP$.

Beyond STEP

The pair-by-pair argument is a first example of a *Markov chain* with absorbing states, and the equation $P = a + sP$ is first-step analysis, the standard way to find absorption probabilities. The same ideas give the probability of winning a whole tennis set or a game of squash, and they lead on to gambler's ruin and random walks. Grimmett and Welsh, *Probability: An Introduction*, develops random walks and Markov chains at exactly this level.

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